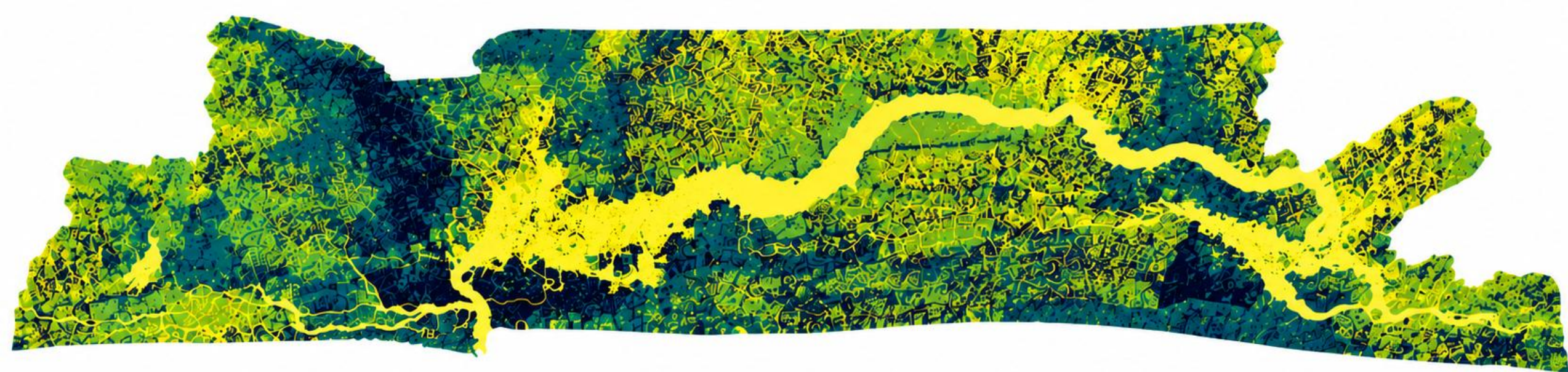




Interpretable Multi-Source Machine Learning for **Urban Classification** in Data-Scarce Regions

Peter Edemewe Ugege • John Aigbokhan • Emmanuel Akalumhe

Forestry Research Institute of Nigeria (FRIN)

Deep Learning Indaba 2026

Research Track | Oral Presentation



DEEP
LEARNING
INDABA
2026

Why Is Accurate Urban Mapping Still So **Difficult**?



Rapid Urban Expansion

Rapid urban growth outpaces mapping capacity.



Highly Heterogeneous Landscapes

Mixture of dense built-up, informal settlements, vegetation, water, and bare surfaces.



Limited High-Quality Training Data

Scarcity of labeled data and inconsistent ground truth reduce model reliability.



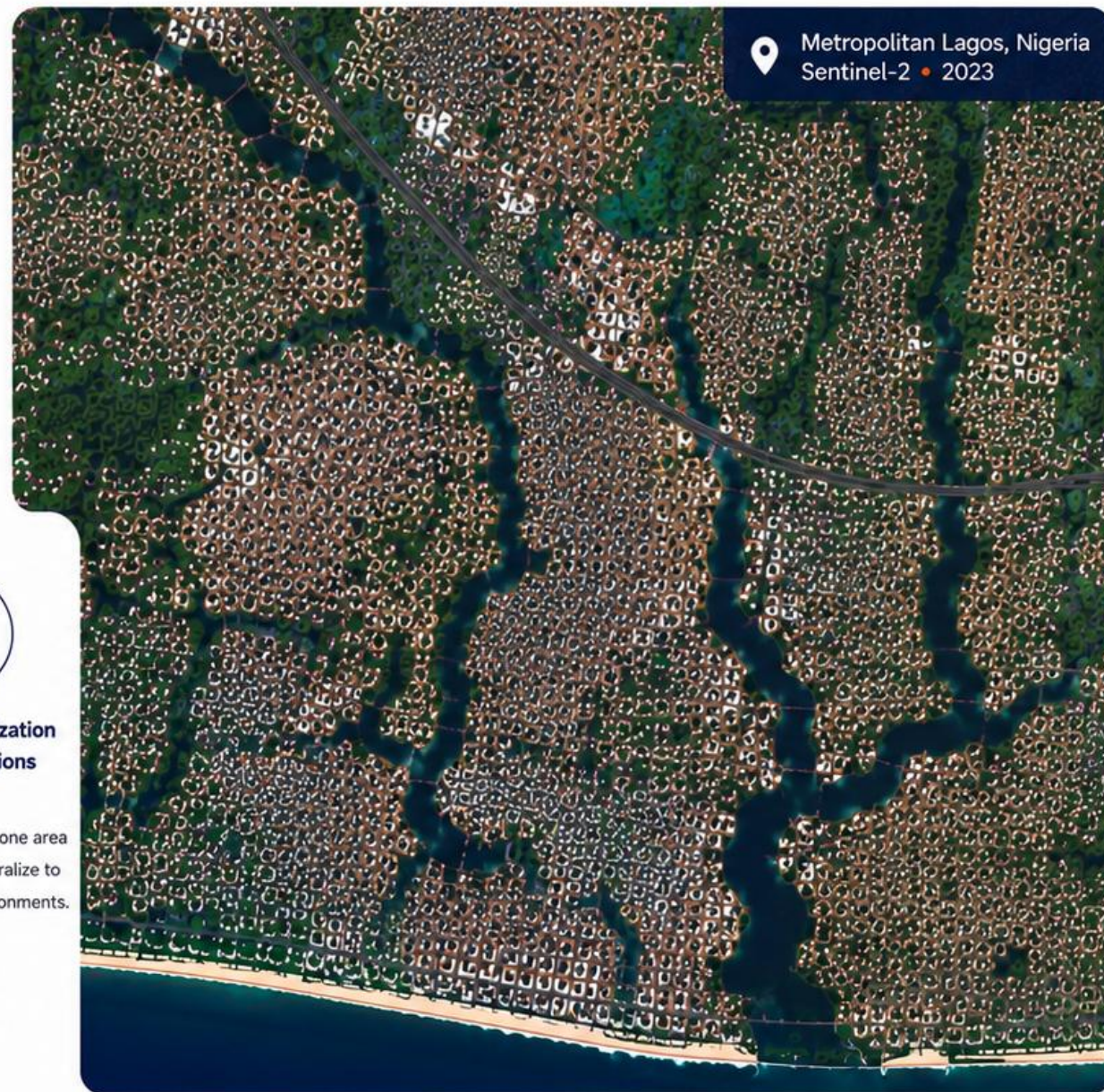
Black-Box Models Are Difficult to Trust

Black-box models lack transparency and actionable insights.



Poor Generalization Across Regions

Models trained in one area often fail to generalize to new, unseen environments.



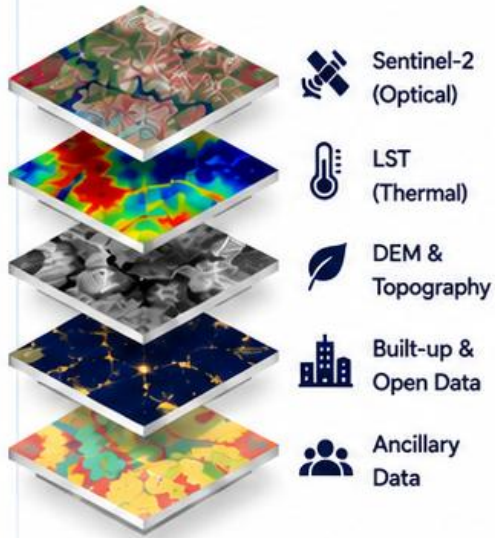
Our Solution: Interpretable Multi-Source Machine Learning.

Our Approach

An interpretable, multi-source machine learning framework for reliable urban classification in data-scarce regions.

1

Multi-Source Data Integration



2

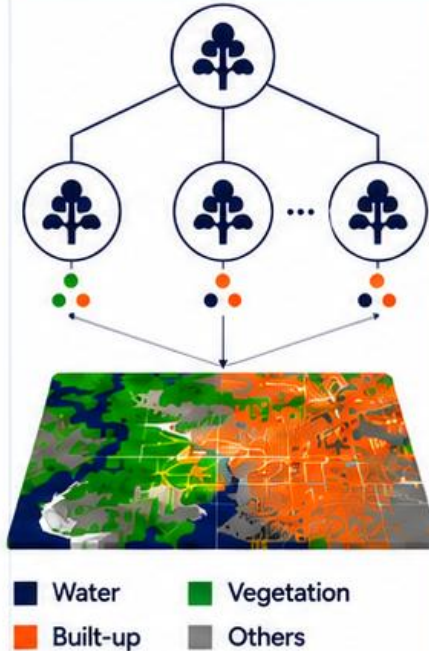
Feature Engineering



- Spectral Indices
- Texture Metrics
- Morphological Features
- Environmental Variables

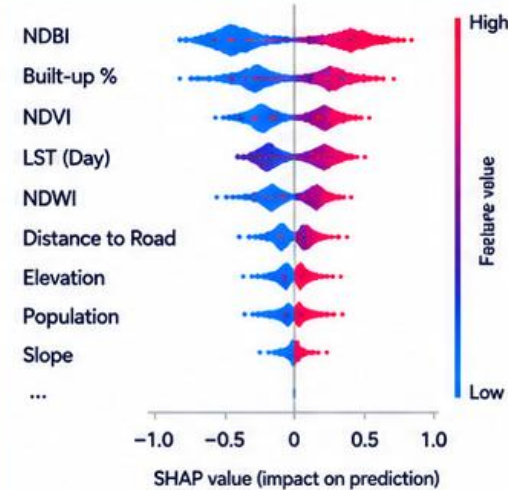
3

Random Forest Classification



4

Explainable AI (SHAP)



Transparent insights into what drives each prediction.

5

Reliable Urban Classification



- ✓ High Accuracy
- ✓ Interpretable Predictions
- ✓ Generalizable Across Regions
- ✓ Supports Data-Driven Decisions



Our Goal: Accurate, explainable, and transferable urban mapping using diverse data and interpretable AI.

Multi-Source Feature Integration

Eleven complementary predictors were integrated into a unified feature space.



Study Area: Lagos State, Nigeria



Feature Groups



Spectral Features

- Blue
- Green
- Red
- NIR
- SWIR
- NDVI



Urban Context

- NDBI
- Population
- Nightlights



Physical Context

- Elevation



Temporal Context

- Year



Feature Matrix

11 predictor variables per sampled location



Random Forest Classifier

Urban / Non-Urban Prediction



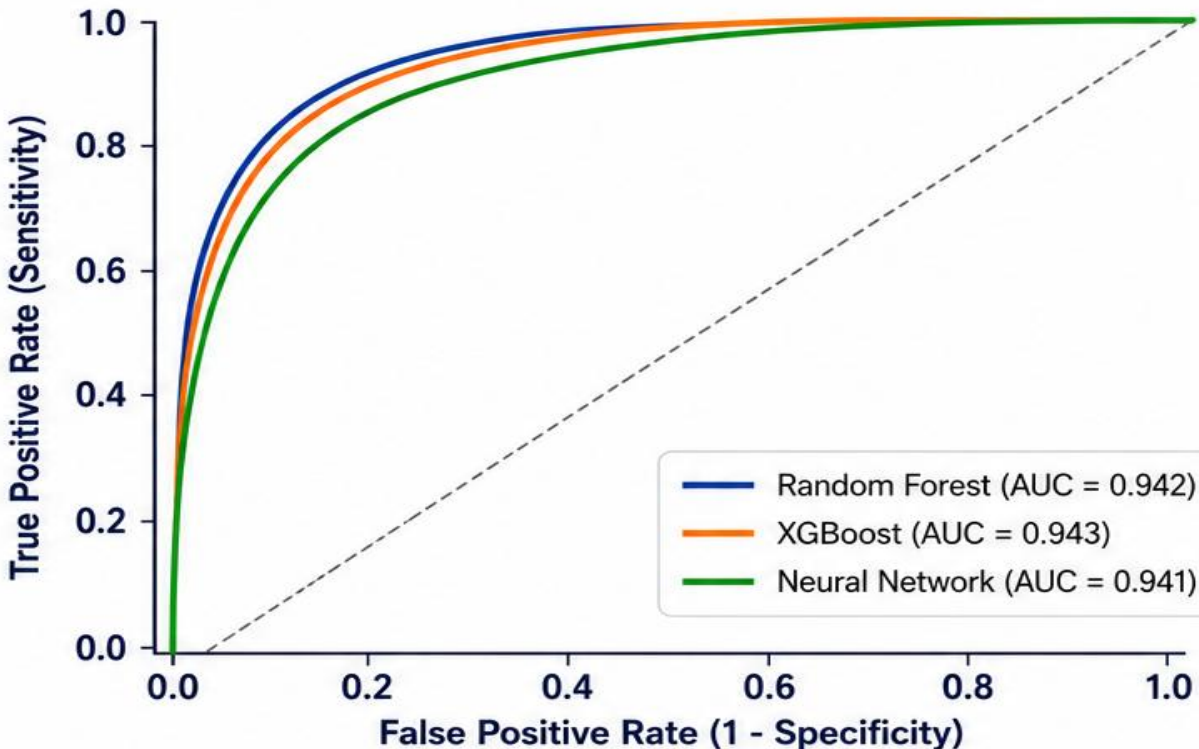
Multi-source feature integration improved **accuracy** and **interpretability**.

Model Performance Evaluation

Three machine learning models demonstrated consistently strong classification performance.



ROC Curve Comparison for Urban Classification



Performance Summary

Model	Accuracy	Precision	Recall	AUC
 Random Forest	0.87	0.87	0.87	0.942
 XGBoost	0.87	0.87	0.87	0.943★
 Neural Network	0.86	0.87	0.86	0.941

★ Best performance



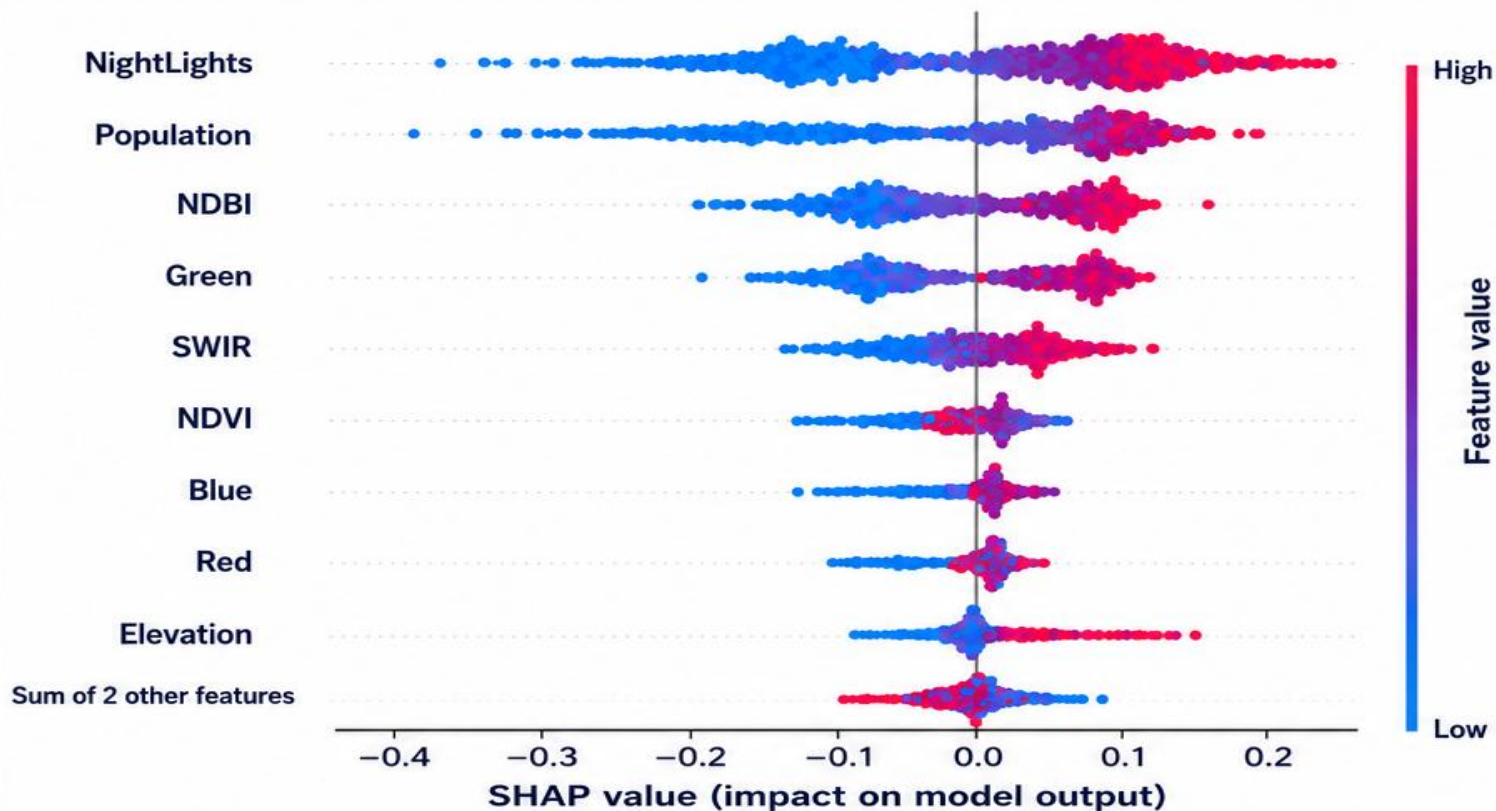
All three models achieved consistently high classification performance, with **AUC values above 0.94**.

Model Explainability with SHAP

SHAP reveals both the importance of each predictor and its influence on urban classification decisions.



SHAP Summary Plot (Impact on Model Output)



Each dot represents a sample. Color shows the feature value (blue = low, red = high). Position shows the impact on prediction (right = increases, left = decreases).



Key Insights



Nightlights

Most influential predictor; high values strongly increase urban probability.



Population

High population consistently drives predictions toward urban areas.



NDBI & Spectral Bands

Provide critical information that refines urban boundaries and built-up patterns.

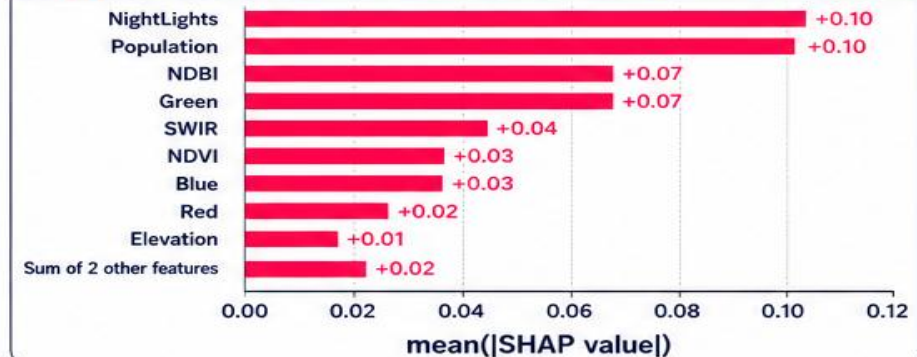


Elevation

Least influential; minimal effect on model decisions in this low-relief area.



Feature Importance Ranking (mean |SHAP value|)



Urban activity indicators (Nightlights, Population, NDBI) are the dominant drivers of classification, while **spectral and terrain variables** provide complementary information.

Key Findings

Three key insights from the proposed framework



1

High Predictive Performance

All three machine learning models achieved consistently strong classification performance (AUC > 0.94).



2

Explainable AI Revealed Meaningful Urban Drivers

Nightlights, Population and NDBI emerged as the dominant drivers of urban classification.



3

Practical Framework for Data-Scarce Regions

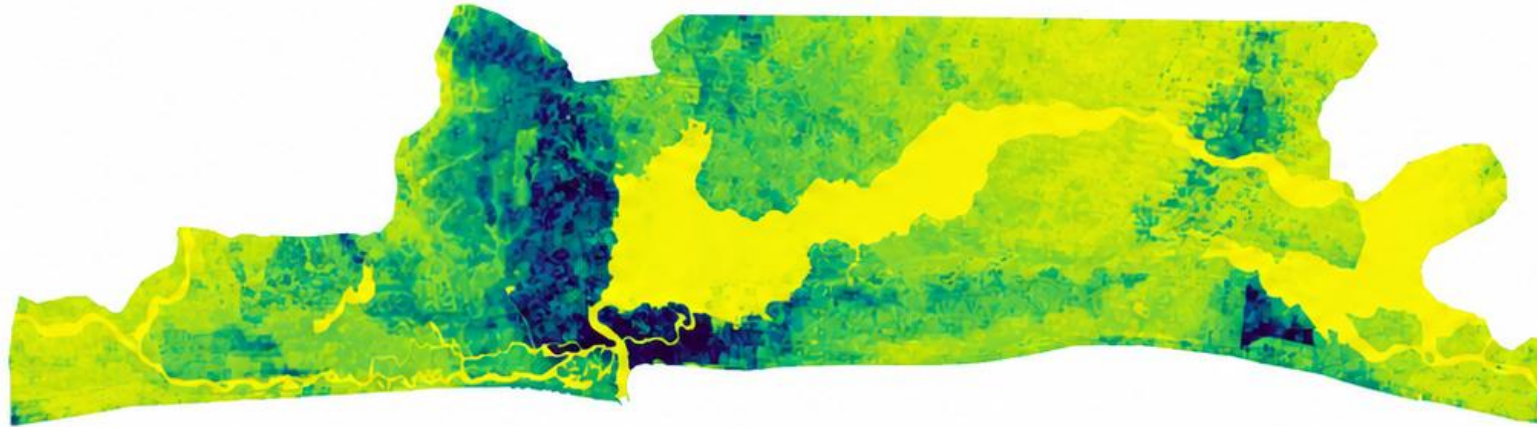
The framework is **transferable** to data-scarce regions using openly available geospatial data.



Multi-source machine learning delivers **accurate**, **interpretable** and **actionable** urban mapping in data-scarce environments.

From Prediction to Decision Support

Accurate, interpretable urban intelligence for data-scarce regions



Urban Probability



Map shows predicted probability of being urban in Lagos, Nigeria (2020) derived from interpretable multi-source machine learning.

Why it matters



Accurate Urban Monitoring

High-resolution, cost-effective mapping enables up-to-date urban monitoring.



Explainable Decision Support

Interpretable AI reveals key urban drivers, building trust for informed decisions.



Transferable Framework

Developed using openly available data, applicable to other data-scarce regions.



Supports Sustainable Planning

Provides evidence for resilient, inclusive and sustainable urban development.



Interpretable multi-source machine learning enables **accurate, transparent** and **scalable** urban mapping for data-scarce regions.



Thank You!

Questions & Discussion



Peter Ugege

Informatics & ICT Resource Centre
Forestry Research Institute of Nigeria (FRIN)



pugege@frin.gov.ng



ORCID: 0000-0002-2722-0076

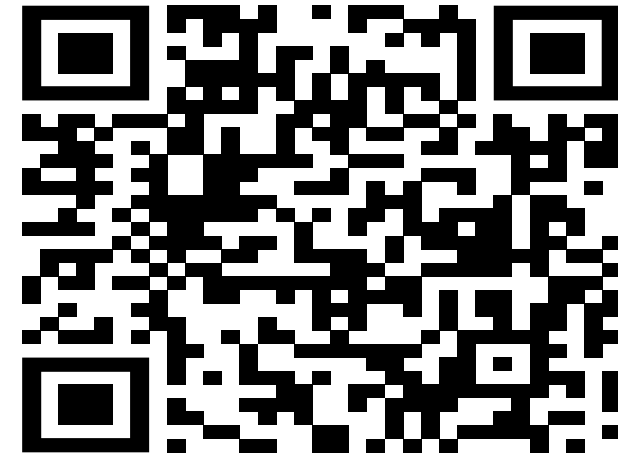


Ibadan, Nigeria



Read the Full Paper

Scan the QR code to read
the full paper.



DLI DEEP LEARNING
INDABA 2026
The African Deep Learning Conference



2 – 7 August 2026



Pan-Atlantic University
Lagos, Nigeria